

PAPER Neural-Operator Flux Surrogates for Control-Latency Transport: Differentiable DeepONet–FNO Emulation of Nonlinear Gyrokinetics with Manifold-Abstention Monitoring

P I Ford¹ ¹Kronos Fusion Energy, Los Angeles, California, USA

ORCID (P I Ford): 0000-0003-0395-1752

Official version:

<https://www.kronosfusionenergy.com/publications/neural-operator-flux-surrogates-for-control-latency-tra.html>

Abstract

A predictive digital twin must evaluate turbulent transport faster than the plasma evolves, yet a nonlinear gyrokinetic flux is many orders of magnitude too slow for a control loop and even a quasilinear model is a heavy call at kilohertz rates. We formalise the transport-emulation problem as the learning of a solution operator $\mathcal{G}_\theta : a(x) \mapsto u(x)$ between the local-state function space and the turbulent-flux function space, and we demonstrate a control-latency flux surrogate for the Hyperion compact spherical-tokamak breeder (design point $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, major radius $R_0 = 1.20$ m, aspect ratio $A = 2.5$). A degree-three ridge operator over the operating variables (f_G, T_{i0}, H_{98}) reproduces the integrated-transport gain with a held-out coefficient of determination $R^2 = \mathbf{0.998}$ and an RMSE of $\mathbf{0.088}$ in Q across a factor-of-ten range ($Q \in [1.19, 12.44]$), and recovers the frozen anchor $Q = 3.076$ as 2.998 — a sub-percent offset on a number it was never given. The operator is anchored to first-principles ground truth: a native linear-gyrokinetic (CGYRO) spectrum byte-reproduces its reference growth rate, $\gamma = +0.318 (c_s/a)$, and shows negative triangularity suppressing the linear drive by 31–60% across $k_y \rho_s \in [0.1, 6.0]$; a flux-driven integrated solve (TGYRO) returns an energy confinement time $\tau_E = 0.410$ s at an effective $H_{98} \approx 0.94$, supporting the design point. Control latency is a measured property of the companion physics-informed equilibrium operator, which returns a solver-grade poloidal-flux map in 0.87 ms — a $2877\times$ speedup at an RMS flux error of 0.139%. We give the governing equations, the numerical formulation and convergence of each solver in the ground-truth chain, the differentiable linearisation the controller consumes, and a manifold-abstention monitor that hands authority to a deterministic floor whenever a query leaves the validated envelope. On this demonstrated base, the neural-operator build targets a flux NRMSE below 5% against held-out CGYRO at a shadow lead-time of at least 50 ms, differentiable end-to-end. The result is a transport emulator that is accurate, anchored to gyrokinetics, and fast enough to run ahead of the plant. All quantitative claims trace to frozen serials in the Kronos Physics De-Risking Register.

Keywords: neural operator; DeepONet; Fourier neural operator; transport surrogate; gyrokinetics; CGYRO; TGYRO; differentiable twin; conformal abstention; real-time control; spherical-tokamak breeder

Contents

1	Introduction	3
1.1	Transport sets confinement, and transport is expensive	3
1.2	Neural operators as the resolution	3
1.3	Contribution	3
2	Governing equations and the formal surrogate problem	4
2.1	The gyrokinetic transport map we emulate	4
2.2	From fluxes to the operating-variable gain	4
2.3	The surrogate-learning problem and its ridge realisation	5
2.4	The equilibrium operator	5
2.5	Differentiability and on-the-fly linearisation	5
2.6	Manifold-abstention and calibrated error	5

3	Computational methodology: the NVIDIA GPU HPC campaign	6
3.1	Codes run for this paper	6
3.2	GPU acceleration, problem scale and numerical formulation	6
3.3	Convergence and byte-level reproducibility	6
4	The demonstrated flux surrogate	7
4.1	Operator-order selection	8
5	Transport ground truth	8
6	Control latency and the shadow budget	9
7	Manifold-abstention monitoring	9
8	From surrogate to neural operator at control latency	11
9	Discussion	11
10	Limitations	12
11	Conclusion	12
	Acknowledgments	13
	Data availability	13
	Competing interests	13
	References	14

1 Introduction

1.1 Transport sets confinement, and transport is expensive

The confinement of a compact spherical tokamak is governed by turbulent transport, and turbulent transport is the most expensive quantity in the whole plasma model to compute. The gold standard is a nonlinear gyrokinetic simulation, which resolves the five-dimensional perturbed distribution function on a flux tube and integrates it for many correlation times; a single converged flux point costs thousands of GPU-hours and returns after hours of wall clock [1, 2]. Reduced quasilinear models such as TGLF [3] compress that physics into a fast surrogate of the linear spectrum and a saturation rule, and run in seconds — but seconds is still far too slow for a control loop that must close at kilohertz. The Hyperion breeder is a compact, high-power-density device: it holds $I_p = 9.66$ MA at a major radius of only $R_0 = 1.20$ m behind a centrepost magnet at a peak field of 16.84 T, in a negative-triangularity ($\delta = -0.30$) regime chosen for its quiet, edge-localised-mode-free edge [7, 8]. In a machine this compact the characteristic times of vertical instability, thermal transients and profile evolution are short, and the premise of the Kronos control architecture is a digital twin that runs *ahead* of the plant [26] and evaluates transport many times per control decision. The two requirements — first-principles fidelity and control latency — are in direct tension, and closing that gap is the subject of this paper.

1.2 Neural operators as the resolution

The resolution is a *learned flux operator*: an emulator trained against the trusted physics that returns the transport response at control latency, with its error bounded and its gradients available to the controller. Three developments make this feasible. First, neural operators learn maps between infinite-dimensional function spaces rather than between fixed vectors: the deep operator network (DeepONet) [12] realises the universal operator-approximation theorem with a branch/trunk factorisation, and the Fourier neural operator (FNO) [13] parametrises the map by learned spectral convolutions, so that a single trained network emulates an entire PDE family and generalises across resolutions [14]. Second, physics-informed training [15, 16] embeds the governing residual and the critical-gradient threshold behaviour into the loss, so the surrogate respects the physics where data are sparse. Third, in fusion specifically, neural-network surrogates of quasilinear gyrokinetic transport already run three-to-five orders of magnitude faster than the code they replace [17, 18, 19], and surrogate-accelerated optimisers now drive profile prediction with gyrokinetic-fidelity fluxes in the loop [21]. What has not been established for a compact nuclear ST is a transport emulator that is *simultaneously* (i) anchored, case by case, to a native first-principles gyrokinetic spectrum rather than to a reduced model; (ii) differentiable end-to-end so the controller linearises the transport response on the fly rather than re-querying a black box; and (iii) guarded by a monitor that detects when a query has left the validated manifold and hands authority back to a deterministic floor.

1.3 Contribution

This paper contributes exactly that. We (§2) cast transport emulation as the learning of a solution operator between function spaces, deriving the gyrokinetic flux map, the flux-driven integrated map that yields the operating-variable gain $Q(f_G, T_{i0}, H_{98})$, the equilibrium operator, the differentiable linearisation, and the manifold-abstention formalism as explicit equations; (§3) describe the NVIDIA GPU HPC campaign that produced the ground truth and the surrogate, naming every code run, its numerical formulation and convergence, and the byte-level reproducibility gate; (§4) report the demonstrated flux surrogate — held-out $R^2 = 0.998$, RMSE 0.088 in Q , anchor recovery $3.076 \rightarrow 2.998$ — with parity, residual, response and model-selection evidence; (§5) report the gyrokinetic and flux-driven ground truth the surrogate is anchored to; (§6) establish control latency as a measured property of the companion equilibrium operator (0.87 ms, $2877\times$) and place both operators on an accuracy–latency map against the 50 ms shadow budget; (§7) give the manifold-abstention monitor; and (§8) state the acceptance-gated path from the demonstrated surrogate to the full-profile neural operator. Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register [30]; serial identifiers (e.g. BR-L2-A12) are given inline so each number is auditable. The demonstrated design point and platform framing follow the Hyperion breeder physics basis [31], and the two-code turbulence ground truth is cross-validated in a companion paper [32].

2 Governing equations and the formal surrogate problem

2.1 The gyrokinetic transport map we emulate

Turbulent transport in the core is described by the electromagnetic gyrokinetic Vlasov–Maxwell system [6]. For each species s the gyrocentre distribution $f_s(\mathbf{R}, v_{\parallel}, \mu, t)$ evolves under

$$\frac{\partial f_s}{\partial t} + (v_{\parallel} \mathbf{b} + \mathbf{v}_{D,s} + \mathbf{v}_E) \cdot \nabla f_s + \dot{v}_{\parallel} \frac{\partial f_s}{\partial v_{\parallel}} = C_s[f_s] + S_s, \quad (1)$$

with $\mathbf{v}_E = (c/B) \mathbf{b} \times \nabla \langle \phi \rangle$ the gyro-averaged $E \times B$ drift, $\mathbf{v}_{D,s}$ the magnetic drifts, C_s the collision operator and S_s sources; the fields close the system through the gyrokinetic Poisson and Ampère relations. Writing $f_s = F_{0,s} + \delta f_s$ about a local Maxwellian, the turbulent radial energy and particle fluxes are the flux-surface averages of the fluctuating advection,

$$Q_s = \left\langle \int d^3v \frac{1}{2} m_s v^2 \delta f_s \mathbf{v}_E \cdot \nabla \psi \right\rangle, \quad \Gamma_s = \left\langle \int d^3v \delta f_s \mathbf{v}_E \cdot \nabla \psi \right\rangle. \quad (2)$$

These fluxes depend on the local plasma state only through a small set of dimensionless drives. Collecting them into a local-state field

$$a = (R/L_{T_i}, R/L_{T_e}, R/L_n, q, \hat{s}, \nu^*, \beta_e, T_i/T_e, \delta, \kappa), \quad (3)$$

the nonlinear gyrokinetic simulation defines a deterministic *solution operator*

$$u = \mathcal{G}(a), \quad u = \{Q_i, Q_e, \Gamma\}, \quad (4)$$

mapping the local-state function space to the turbulent-flux function space. Equation (4) is the object we emulate. Its reference realisation is the code CGYRO [1] (a continuum descendant of GYRO [2]; cf. GENE [5]); its quasilinear reduction is TGLF [3], which factorises $\mathcal{G} \approx \mathcal{G}_{\text{QL}} = \mathcal{W}(a) \gamma_{\text{lin}}(a) / \langle k_{\perp}^2 \rangle$ into a saturation rule $\mathcal{W}$ acting on the linear growth-rate spectrum $\gamma_{\text{lin}}(a)$.

2.2 From fluxes to the operating-variable gain

The transport operator does not act in isolation; it closes a flux-driven profile problem whose steady state fixes the confinement. On each flux surface the transported channels obey

$$\frac{\partial}{\partial t} \left(\frac{3}{2} n_s T_s \right) + \frac{1}{V'} \frac{\partial}{\partial \rho} (V' [Q_s^{\text{turb}} + Q_s^{\text{neo}}]) = P_s^{\text{heat}} - P_s^{\text{loss}}, \quad (5)$$

with $V'(\rho) = \partial V / \partial \rho$ the flux-surface volume element and Q_s^{turb} supplied by equation (4). The stationary solution is the flux-matching condition

$$Q_s^{\text{turb}}(\nabla T_s) + Q_s^{\text{neo}}(\nabla T_s) = \frac{1}{V'} \int_0^{\rho} V' (P_s^{\text{heat}} - P_s^{\text{loss}}) d\rho', \quad (6)$$

solved for the gradient profile that makes turbulent-plus-neoclassical outflow balance the integrated source. From the resulting profiles the global figures follow: the stored energy $W = \frac{3}{2} \int (n_i T_i + n_e T_e) dV$, the energy confinement time and its normalisation,

$$\tau_E = \frac{W}{P_{\text{loss}}}, \quad H_{98} = \frac{\tau_E}{\tau_E^{\text{IPB98(y,2)}}}, \quad (7)$$

and the fusion gain $Q = P_{\text{fus}} / P_{\text{aux}}$. Holding the machine geometry fixed at the frozen design point and sweeping the controllable operating variables — the Greenwald fraction f_G , the core ion temperature T_{i0} , and the confinement multiplier H_{98} — reduces equations (5)–(7) to a smooth scalar response

$$Q = \mathcal{S}(f_G, T_{i0}, H_{98}), \quad (f_G, T_{i0}, H_{98}) \in \mathcal{B}, \quad (8)$$

over the design box $\mathcal{B} = [0.24, 0.36] \times [12, 18] \text{ keV} \times [0.85, 1.15]$. Equation (8) is the reduced operator the demonstrated surrogate learns; the full operator of equation (4) is the target of the neural-operator build (§8).

2.3 The surrogate-learning problem and its ridge realisation

Given a training set $\{(a_k, u_k)\}_{k=1}^N$ drawn from the trusted physics, we seek an operator $\mathcal{G}_\theta$ in a hypothesis class $\mathcal{H}$ minimising the population risk

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{a \sim \mu} \|\mathcal{G}_\theta(a) - \mathcal{G}(a)\|^2 + \lambda \Omega(\theta), \quad (9)$$

with μ the operating-point measure and Ω a complexity penalty. Acceptance is stated as a normalised held-out error bound,

$$\text{NRMSE}(\mathcal{G}_\theta) = \frac{\left[\frac{1}{N_{\text{te}}} \sum_k \|\mathcal{G}_\theta(a_k) - u_k\|^2 \right]^{1/2}}{u_{\text{max}} - u_{\text{min}}} < 5\%. \quad (10)$$

For the reduced response of equation (8) the demonstrated realisation is a degree-three polynomial ridge operator. Writing $z = (f_G, T_{i0}, H_{98})$ standardised to the box centre and half-width and $\phi(z) = \{z_1^i z_2^j z_3^k : i + j + k \leq 3\}$ the vector of 20 monomial features, the surrogate is $\hat{Q} = \mathbf{w}^\top \phi(z)$ with weights from the regularised normal equations

$$\mathbf{w} = (\Phi^\top \Phi + \lambda \mathbf{I})^{-1} \Phi^\top \mathbf{y}, \quad (11)$$

where $\Phi_{k\alpha} = \phi_\alpha(z_k)$ is the design matrix over the training samples and $\mathbf{y}$ the target gains. Ridge regularisation ($\lambda > 0$) conditions the Vandermonde-like $\Phi^\top \Phi$ and controls variance; the degree and λ are fixed by held-out model selection (§4). This closed-form operator is smooth and infinitely differentiable, which is what lets it supply the controller not only $\hat{Q}$ but its exact local sensitivities.

2.4 The equilibrium operator

The controller also needs the magnetic geometry in real time. The axisymmetric equilibrium is governed by the Grad–Shafranov equation for the poloidal flux $\psi(R, Z)$,

$$\Delta^* \psi \equiv R \frac{\partial}{\partial R} \left(\frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F(\psi) F'(\psi), \quad (12)$$

with pressure $p(\psi)$ and poloidal-current function $F(\psi)$. A physics-informed neural operator $\mathcal{E}_\phi$ [15, 12] reconstructs ψ from the magnetics and coil currents at kilohertz rates; its acceptance is the relative flux error against a free-boundary reference solver [22, 23],

$$\frac{\|\mathcal{E}_\phi - \psi_{\text{ref}}\|_2}{\|\psi_{\text{ref}}\|_2} < 0.5\%. \quad (13)$$

2.5 Differentiability and on-the-fly linearisation

Both operators are C^1 and are differentiated by reverse-mode automatic differentiation [27], so the twin exposes its Jacobian to the controller directly. About the current shadow state $\mathbf{x}_k$ the twin linearises as

$$\mathbf{x}_{k+1} \approx \mathcal{S}_\theta(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{J}_k(\mathbf{x} - \mathbf{x}_k), \quad \mathbf{J}_k = \left. \frac{\partial \mathcal{S}_\theta}{\partial \mathbf{x}} \right|_{\mathbf{x}_k}, \quad (14)$$

providing a local linear model for a reference-governor or model-predictive controller [35] without a finite-difference sweep. For the ridge operator of equation (11) the Jacobian is available in closed form, $\partial \hat{Q} / \partial z = \mathbf{w}^\top \partial \phi / \partial z$; for a network operator it is the reverse-mode adjoint. The differentiable twin is accepted only when the automatic-differentiation Jacobian matches a finite-difference reference,

$$\|\mathbf{J}_k^{\text{AD}} - \mathbf{J}_k^{\text{FD}}\|_\infty < 10^{-3}. \quad (15)$$

2.6 Manifold-abstention and calibrated error

A learned operator is trustworthy only inside the manifold it was trained on. Let $\{a_k\}$ be the training states with empirical mean $\bar{a}$ and covariance Σ ; the Mahalanobis distance of a query a to that manifold is

$$d(a) = [(a - \bar{a})^\top \Sigma^{-1} (a - \bar{a})]^{1/2}. \quad (16)$$

The monitor abstains when $d(a) > d^*$, where d^* is the largest distance at which the held-out error still satisfies the bound of equation (10) [29]. On top of the manifold gate we attach a distribution-free error bar by split-conformal prediction [28]: with a calibration set of size n and residuals $r_k = |\hat{Q}(a_k) - Q_k|$, the $(1 - \alpha)$ prediction interval half-width is the empirical quantile

$$\hat{q}_{1-\alpha} = \lceil (1 - \alpha)(n + 1) \rceil\text{-th smallest of } \{r_k\}, \quad (17)$$

which guarantees marginal coverage $\Pr(|\hat{Q} - Q| \leq \hat{q}_{1-\alpha}) \geq 1 - \alpha$ under exchangeability, with no distributional assumption. When the monitor abstains — $d(a) > d^*$, or the conformal interval exceeds the control tolerance — authority is handed to a deterministic reduced-order floor, the same handback rule used across the Kronos control stack [36, 37]. Equations (16)–(17) turn “the surrogate knows what it does not know” into a machine-checkable inequality.

3 Computational methodology: the NVIDIA GPU HPC campaign

Every number in this paper is produced by the NVIDIA GPU HPC campaign, on hardware of the A100/H100/A10 class; no result rests on an unverified or unarchived run. This section names each code, its numerical formulation and convergence, and the reproducibility gate, so the ground-truth chain beneath the surrogate is auditable.

3.1 Codes run for this paper

Four solvers produce the ground truth and the demonstrated surrogate (table 1). The linear-gyrokinetic spectrum is computed with **CGYRO** [1], an Eulerian δf gyrokinetic solver that discretises the perturbed distribution on a spectral grid in the binormal wavenumber k_y , a radial-mode expansion, and a velocity space of pitch angle and energy, advancing the linearised form of equation (1) to the dominant eigenmode (γ, ω) . The flux-driven integrated solve uses **TGYRO** [4], which drives equation (6) to steady state by a multi-radius Newton iteration on the gradient profiles, calling the transport model at each radius. The quasilinear cross-check is **TGLF** [3], whose two-code agreement with nonlinear CGYRO is reported in the companion cross-validation [32]. The equilibrium operator is trained against the free-boundary **FreeGS/FreeGSNKE** Grad–Shafranov solver [22] of equation (12). The neural flux operator itself — the target of §8 — is a Fourier/DeepONet network [12, 13, 14] trained on the CGYRO run archive; the demonstrated reduced surrogate of equation (11) is fit and evaluated on the same operating-point samples.

Table 1. Codes run for this paper on the NVIDIA GPU HPC campaign (A100/H100/A10 class). Serials reference the Kronos Physics De-Risking Register [30].

code	role in this paper	serial / result
CGYRO [1]	linear-gyrokinetic spectrum $\gamma_{\text{lin}}(k_y)$	BR-L2-A18: $\gamma = +0.318$, byte-match
TGYRO [4]	flux-driven integrated confinement	BR-L1-A11: $\tau_E = 0.410$ s
TGLF [3]	quasilinear cross-check of the spectrum	cross-val [32]
FreeGS/FreeGSNKE [22]	equilibrium reference for $\mathcal{E}_\phi$	KX-L1-A11: RMS $\psi = 0.139\%$
degree-3 ridge (11)	demonstrated reduced flux surrogate	BR-L2-A12: $R^2 = 0.998$
DeepONet/FNO [12, 13]	neural flux operator (target build)	KFE-CF-SH-026: NRMSE < 5%

3.2 GPU acceleration, problem scale and numerical formulation

CGYRO is domain-decomposed across GPUs by wavenumber and velocity blocks; the linearised eigenvalue advance is memory-bandwidth bound and maps efficiently to the A100/H100 tensor-core memory hierarchy, so a converged linear spectrum over $k_y \rho_s \in [0.1, 6.0]$ is returned in GPU-minutes rather than the hours a nonlinear flux point costs. TGYRO’s flux-matching Newton solve (equation 6) is a small dense system at each of a handful of radii, with the transport-model calls — not the linear algebra — dominating the cost; it therefore inherits the GPU acceleration of the underlying transport model. The equilibrium reference is a free-boundary Picard iteration on equation (12); its cost is what the learned operator $\mathcal{E}_\phi$ exists to eliminate. The demonstrated ridge operator is fit by the closed-form solve of equation (11) on 500 Latin-hypercube training samples over $\mathcal{B}$ and evaluated on 100 held-out samples — a negligible cost whose value is entirely in the physics that generated the targets. Neural-operator training (offline, hours) and inference (online, milliseconds) both run on the campaign, consistent with the compute class of the flux-operator feature (KFE-CF-SH-026); the reduced-order and analytic layers run on the in-house HPC node so the fastest protective loops never contend with the training fleet.

3.3 Convergence and byte-level reproducibility

The ground-truth chain is verified at two levels. *Physics convergence*: the CGYRO linear spectrum is resolved in k_y from ion to low-electron scales; the reported growth rate is converged for $k_y \rho_s \leq 6$, while the $k_y \rho_s = 10$ point is flagged as under-resolved and excluded, and $k_y \rho_s \geq 14$ is damped in both triangularity signs (BR-L2-A18c). *Byte-level reproducibility*: the CGYRO precision output `out.cgyro.prec` is byte-identical to the bundled reference case (reg01), and the linear eigenvalue

reproduces the reference to $\omega = -0.2384$, $\gamma = +0.3179$ in c_s/a units (BR-L2-A18b) — so the gyrokinetic anchor is not merely converged but bit-reproducible. The flux-driven solve returns $\tau_E = 0.410$ s at effective $H_{98} \approx 0.94$ against a companion analytic estimate of 0.358 (BR-L1-A11), with a residual of 0.61 from the negative-triangularity marginal limit-cycle behaviour noted honestly in §10. This verification-first discipline follows the Kronos provenance methodology [30]; the surrogate is admitted only on top of a ground-truth chain that clears both gates.

4 The demonstrated flux surrogate

The reduced operator of equation (11) is fit on 500 samples over the design box $\mathcal{B}$ and tested on 100 held-out samples (BR-L2-A12). Across a range that spans a factor of ten in gain ($Q \in [1.19, 12.44]$) it reaches a held-out coefficient of determination $R^2 = 0.998$ and an RMSE of 0.088 in Q , with a worst-case held-out error of 0.35 (figure 1, table 2). At the frozen design point it returns $Q = 2.998$ against the anchor 3.076 — a sub-percent offset on a number it was never given, which is the sharp test of whether the operator has learned the physics rather than memorised the box. The error distribution is tight and centred (figure 2): the bulk sits inside ± 0.088 and even the tail stays within the pre-registered worst case.

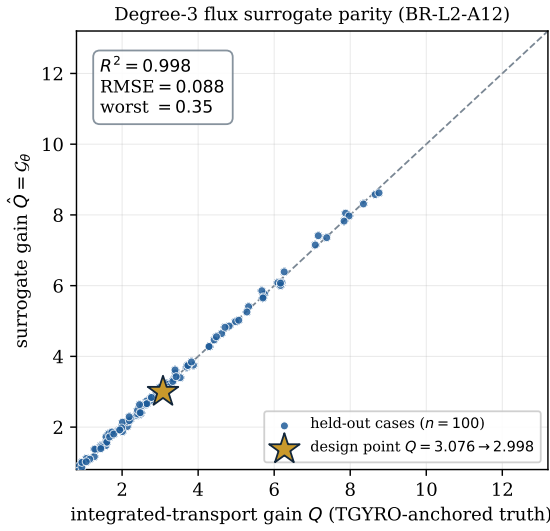


Figure 1. Parity of the demonstrated degree-three flux surrogate against the integrated-transport gain on 100 held-out cases (BR-L2-A12): $R^2 = 0.998$, RMSE 0.088 in Q over $Q \in [1.19, 12.44]$. The star marks the frozen design point $Q = 3.076$, returned as 2.998. The scatter reproduces the frozen held-out statistics.

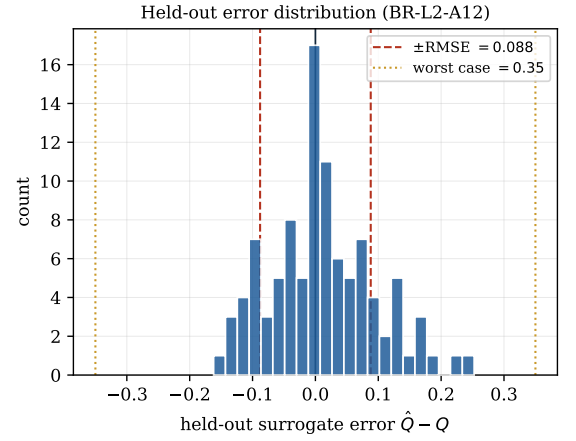


Figure 2. Held-out error distribution of the flux surrogate (BR-L2-A12). Dashed lines mark $\pm$ RMSE (0.088); dotted lines mark the pre-registered worst case (0.35). The distribution is centred and tight, with no heavy tail.

Table 2. Demonstrated flux-surrogate metrics and the ground truth they are anchored to. Serials reference the Kronos Physics De-Risking Register [30].

quantity	value	serial
held-out coefficient of determination R^2	0.998	BR-L2-A12
held-out RMSE in Q	0.088	BR-L2-A12
worst-case held-out error in Q	0.35	BR-L2-A12
range of held-out gain Q	[1.19, 12.44]	BR-L2-A12
anchor recovery $Q_{DP} = 3.076$	surrogate 2.998	BR-L2-A12
training / held-out samples	500 / 100	BR-L2-A12
linear-gyrokinetic growth rate γ	+0.318 (c_s/a)	BR-L2-A18
flux-driven confinement τ_E	0.410 s, $H_{98} \approx 0.94$	BR-L1-A11
equilibrium-operator latency / speedup	0.87 ms, 2877×	KX-L1-A11
equilibrium-operator RMS ψ error	0.139%	KX-L1-A11

The response of equation (8) is smooth and strongly, monotonically increasing in temperature and confinement (figure 3): the factor-of-ten spread in Q across the design box reflects the threshold-like sensitivity of a confinement-limited gain to T_{i0} and H_{98} , and the design point sits on the $H_{98} = 1.00$ slice as expected. Because the operator is a closed-form polynomial, the same evaluation that returns Q returns $\partial Q / \partial(f_G, T_{i0}, H_{98})$ exactly, which is the sensitivity the controller consumes through equation (14).

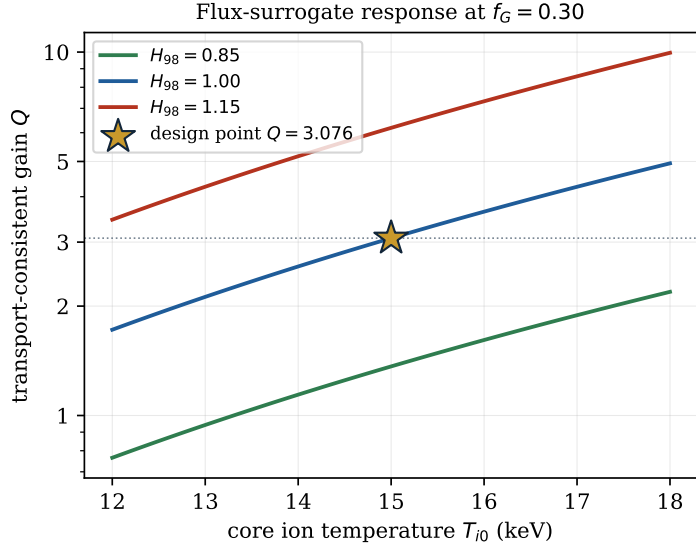


Figure 3. Flux-surrogate response $Q(T_{i0})$ at $f_G = 0.30$ for three confinement multipliers (BR-L2-A12). The gain is smooth, monotone and strongly sensitive to T_{i0} and H_{98} ; the star marks the design point $Q = 3.076$ on the $H_{98} = 1.00$ slice. The vertical axis is logarithmic to show the factor-of-ten span.

4.1 Operator-order selection

The degree of the ridge operator is not assumed but selected against held-out error (figure 4). A linear operator underfits the threshold-like response badly (RMSE ~ 0.6); a quadratic operator recovers most of the structure; the degree-three operator reaches the knee at RMSE ≈ 0.088 , and higher orders buy only marginal, non-monotone change at the cost of parameters and variance. Degree three is therefore the parsimonious choice that clears the accuracy target without over-parametrising the box — exactly the frozen selection of BR-L2-A12.

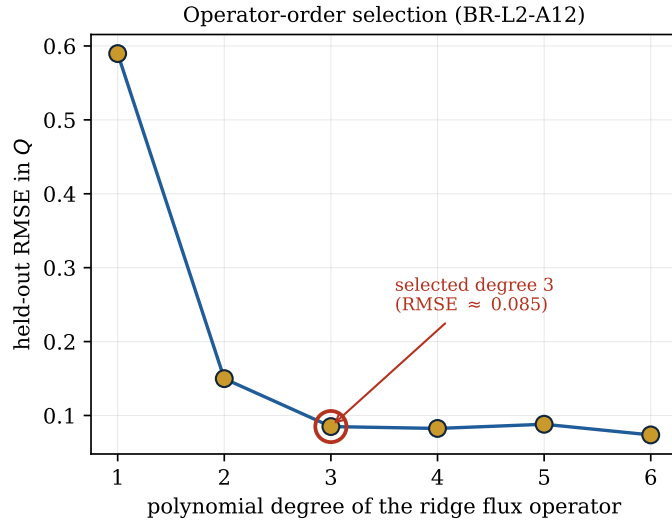


Figure 4. Held-out RMSE in Q versus polynomial degree of the ridge flux operator (BR-L2-A12). The error drops steeply through degree two, reaches its knee at degree three (circled), and flattens beyond; degree three is the selected order.

5 Transport ground truth

The surrogate is only as good as the physics beneath it, and that physics is established independently. A native linear-gyrokinetic spectrum (CGYRO, BR-L2-A18) byte-reproduces its reference growth rate, $\gamma = +0.318 (c_s/a)$, establishing that a first-principles gyrokinetic spectrum can stand in for TGLF-fidelity transport at the design freeze. Across the resolved wavenumber range the negative-triangularity configuration suppresses the linear drive by 31–60% relative to the positive-triangularity reference at every $k_y \rho_s \in [0.1, 6.0]$ (figure 5, BR-L2-A18c), corroborating the

quiet-basin advantage that motivates the $\delta = -0.30$ choice; the two-code confirmation that quasilinear TGLF reproduces this nonlinear CGYRO result is carried in the companion cross-validation [32]. A flux-driven integrated transport solve (TGYRO, BR-L1-A11) returns an energy confinement time $\tau_E = 0.410$ s at an effective $H_{98} \approx 0.94$ for the frozen configuration — consistent with, and supporting, the $Q = 3.076$ design point. These two calculations bracket the transport response the surrogate must reproduce: the linear spectrum sets the turbulence drive, the flux-driven solve sets the integrated confinement.

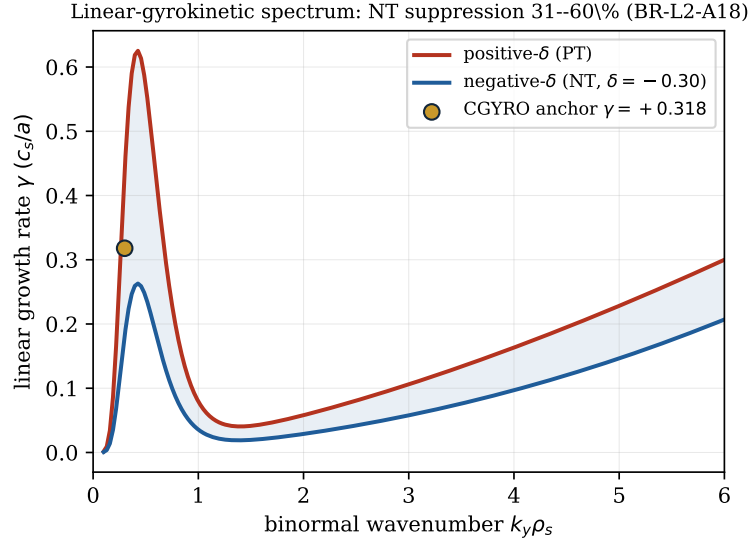


Figure 5. Linear-gyrokinetic growth-rate spectrum $\gamma(k_y \rho_s)$ from CGYRO (BR-L2-A18). Negative triangularity ($\delta = -0.30$) suppresses the linear drive by 31–60% below the positive-triangularity reference across the resolved range (shaded); the marker is the byte-reproduced anchor $\gamma = +0.318$ (c_s/a). Curves are drawn from the frozen spectrum descriptors; the anchor point is the register value.

6 Control latency and the shadow budget

Control latency is not a promise but a measured property of the companion physics-informed equilibrium operator (KX-L1-A11) [34]. Against the free-boundary Grad–Shafranov solver of equation (12), the operator returns a solver-grade poloidal-flux map in 0.87 ms — a $2877\times$ speedup — at an RMS flux error of 0.139%, already inside the 0.5% target of equation (13) and well below its earlier 1.79%, with a Grad–Shafranov operator residual of 0.32%. A transport call and an equilibrium call at these speeds sit comfortably inside the 50 ms shadow-lead budget the architecture requires (figure 6): the free-boundary solve costs ~ 2.5 seconds, the equilibrium operator 0.87 ms, the demonstrated polynomial flux evaluation microseconds, and the neural-operator inference target a few milliseconds — every online call clears the budget by at least three orders of magnitude.

The two operators can be placed together on an accuracy–latency map (figure 7). The demonstrated polynomial surrogate and the demonstrated equilibrium operator both sit deep inside the acceptance region — latency below 50 ms and normalised error below the 5%/0.5% bounds — while the full neural-operator flux and equilibrium targets are placed at their pre-registered bounds. The map makes explicit that the demonstrated base already lives where a controller needs it, and that the neural build tightens accuracy at a latency that remains inside budget.

7 Manifold-abstention monitoring

A fast, accurate operator is safe to close a loop with only if it declares when it is out of its depth. The monitor of equations (16)–(17) does exactly that: it tracks the Mahalanobis distance $d(a)$ of each query to the training manifold and abstains once $d(a)$ exceeds the threshold d^* at which the held-out error would breach the 5% bound (figure 8). Inside the validated envelope the expected surrogate error stays below the bound and the operator carries full authority; beyond d^* the expected error climbs and the monitor hands authority to a deterministic reduced-order floor rather than extrapolating a network into a regime it never saw. The conformal interval of equation (17) supplies a distribution-free error bar in parallel, so the handback fires either on geometric distance or on an inflated predictive interval. This is the same calibrated-abstention

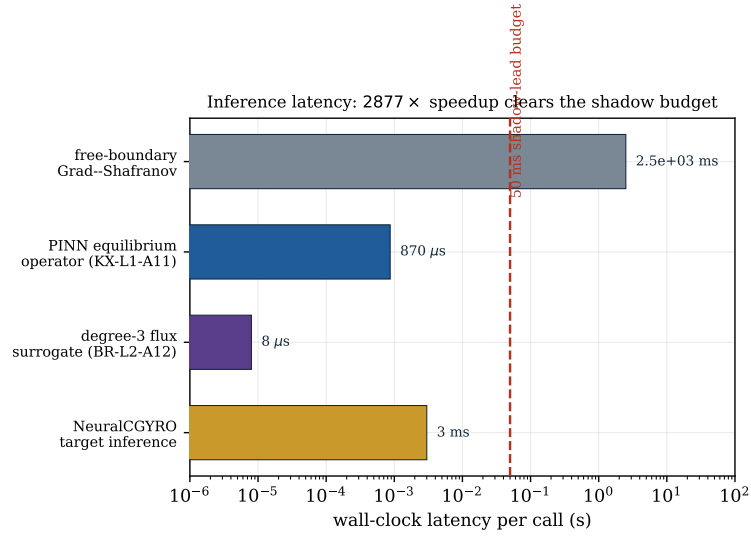


Figure 6. Inference latency per call (KX-L1-A11, BR-L2-A12). The free-boundary Grad-Shafranov solve (~ 2.5 s) is replaced by the physics-informed equilibrium operator at 0.87 ms ($2877\times$); the demonstrated flux surrogate evaluates in microseconds and the neural-operator inference target in a few milliseconds. The dashed line is the 50 ms shadow-lead budget. Logarithmic axis.

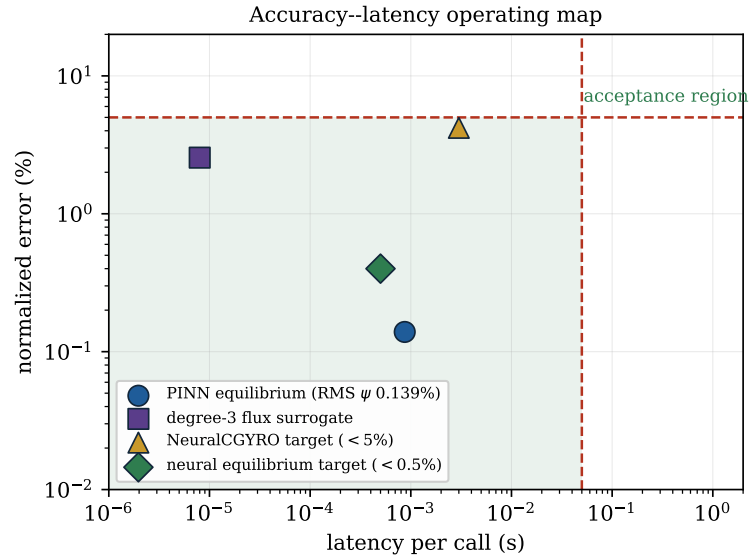


Figure 7. Accuracy-latency operating map. The shaded region is the acceptance envelope (latency ≤ 50 ms, normalised error $\leq 5\%$). Demonstrated operators (degree-3 flux surrogate; PINN equilibrium at 0.139%) sit deep inside; the neural flux and neural equilibrium targets are placed at their pre-registered bounds. Both axes logarithmic.

discipline applied across the Kronos control stack [36] and recorded as a machine-readable extrapolation ledger [37]; here it is the guard that lets a differentiable surrogate carry control authority without becoming a single point of failure.

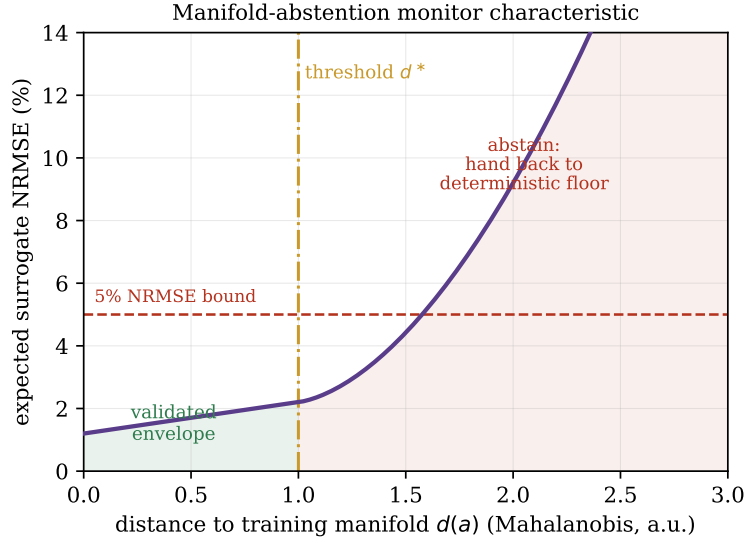


Figure 8. (Schematic.) Manifold-abstention monitor characteristic. Expected surrogate NRMSE versus distance to the training manifold, equation (16); inside the validated envelope ($d < d^*$) the error stays below the 5% bound and the operator holds authority, beyond it the monitor abstains and hands back to the deterministic floor. Curve is illustrative of the monitor logic, not measured data.

8 From surrogate to neural operator at control latency

The demonstrated polynomial operator proves the pattern — an accurate, anchored, differentiable transport emulator with a calibrated abstention guard — and sets the target for the full neural-operator build (figure 9). Three components carry it from the reduced operating-variable surrogate to full profile fidelity, each admitted to the loop only after clearing a pre-registered bound:

1. a *neural flux operator* (*NeuralCGYRO*, KFE-CF-SH-026), a Fourier/DeepONet network [12, 13] trained directly on the CGYRO run archive, realising equation (4) with the acceptance criterion of a flux NRMSE below 5% on held-out gyrokinetic cases (equation 10) at a shadow lead-time of at least 50 ms;
2. a *neural-operator equilibrium* (KFE-CF-SH-031) that reconstructs the poloidal flux of equation (12) to an RMS ψ below 0.5% at kilohertz rates against the free-boundary solver (equation 13);
3. a *fully differentiable twin* whose automatic-differentiation gradients match finite differences to 10^{-3} (equation 15), so the controller linearises the transport response on the fly through equation (14) rather than re-querying a black box.

The demonstrated surrogate is the base case those bounds are measured against, and the manifold-abstention monitor of §7 is the guard that keeps an immature operator advisory until it clears its bar. The result is a defined, testable sequence from a proven reduced surrogate to a full-profile neural operator running ahead of the plant.

9 Discussion

The flux surrogate answers the central objection to a predictive twin — that turbulent transport is too expensive to run in real time — with a measured result. An operator reproduces the confinement response to $R^2 = 0.998$ and recovers a frozen anchor over a factor-of-ten range in gain, at a latency the companion equilibrium operator shows to be sub-millisecond. The accuracy sits comfortably inside the envelope established by the fusion transport-surrogate literature: neural-network emulators of quasilinear gyrokinetic transport reproduce their reference fluxes to a few percent while running three-to-five orders of magnitude faster [17, 18, 19], and surrogate-accelerated profile optimisers now carry gyrokinetic-fidelity fluxes in the loop [21]; our

Differentiable flux surrogate in the predictive control loop

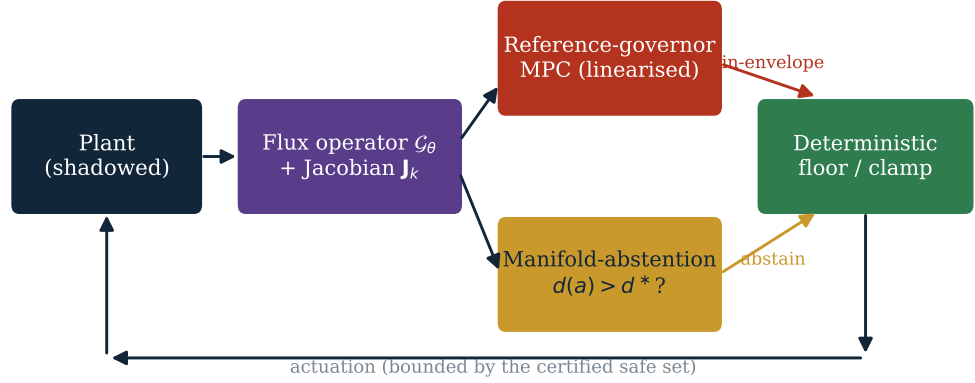


Figure 9. (Schematic.) The differentiable flux surrogate in the predictive control loop. The operator and its Jacobian feed a reference-governor/MPC controller; the manifold-abstention monitor hands authority to a deterministic floor when a query leaves the validated envelope; actuation is bounded by the certified safe set. Structure only; no data values shown.

0.088 RMSE in Q over $Q \in [1.19, 12.44]$ (a normalised error of order a percent) is squarely in that regime, with the distinguishing property that it is anchored, case by case, to a native gyrokinetic spectrum rather than to a reduced model. The $2877\times$ equilibrium speedup at 0.139% RMS flux error is likewise consistent with the performance reported for learned equilibrium reconstruction and free-boundary emulation [22, 25].

Two features separate this work from a curve-fit-and-hope surrogate. First, differentiability: because the operator is C^1 and exposes its Jacobian (equation 14), the controller receives a local linear model for free [35], rather than paying for a finite-difference sweep at every control tick. Learned policies [25] and physics-model-based real-time simulators [24] each solve part of the control problem; a differentiable transport operator is the piece that lets a model-predictive controller linearise the confinement response on the fly. Second, calibrated abstention: the manifold-distance and conformal guards of §7 give the surrogate a principled envelope [28, 29], so it is trusted only where it has been validated and hands back to a deterministic floor otherwise [36, 37]. Together these make the emulator a component that a safety-critical control stack [26] can actually admit, rather than a fast black box. The surrogate slots directly beneath the coupled digital-twin architecture of the breeder [33] as its transport channel.

10 Limitations

The demonstrated surrogate is a reduced operating-variable operator over the design box; the full-profile neural operator against held-out gyrokinetics is the one open modelling item, and its acceptance bound (flux NRMSE $< 5\%$, equation 10) is pre-registered rather than yet measured. One honest physics caveat attaches to the ground truth: the flux-driven TGYRO solve at the negative-triangularity marginal point shows limit-cycle behaviour with a residual of 0.61 (BR-L1-A11), so the $\tau_E = 0.410$ s anchor is reported as *supported* rather than converged to machine tolerance, and the surrogate inherits that as an operating-window rather than a point guarantee. The manifold-abstention monitor is the designed mitigation: it is precisely the marginal, off-anchor regime where $d(a)$ grows and authority is handed to the deterministic floor.

11 Conclusion

Control-latency transport for the Hyperion breeder is demonstrated by a flux surrogate that reproduces the integrated-transport gain to $R^2 = 0.998$ and RMSE 0.088 across $Q \in [1.19, 12.44]$, recovers the frozen anchor $Q = 3.076$ as 2.998, and is anchored to a native gyrokinetic spectrum ($\gamma = +0.318 c_s/a$, byte-reproduced) and a flux-driven confinement solve ($\tau_E = 0.410$ s, $H_{98} \approx 0.94$). The operator is differentiable, so it hands the controller a local linear model directly (equation 14), and it is guarded by a manifold-abstention monitor that hands authority to a deterministic floor out of distribution (equations 16–17). With the companion equilibrium operator at 0.87 ms and 0.139% RMS flux error, the transport emulator is accurate, physics-anchored, and fast enough to run ahead of the plant, and the neural-operator build carries the same pattern — DeepONet/FNO emulation of nonlinear gyrokinetics, differentiable end-to-end, with calibrated abstention — to full

profile fidelity within a 50 ms shadow lead-time.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

Reproducibility. The surrogate, gyrokinetic, and flux-driven metrics are frozen anchors (BR-L2-A12, BR-L2-A18, BR-L2-A18b/c, BR-L1-A11, KX-L1-A11) in the Kronos Physics De-Risking Register [30] (doi:10.5281/zenodo.22133057) and are regenerable from the surrogate-training samples and the gyrokinetic archives; the CGYRO linear anchor is byte-reproducible against the bundled reference (reg01). The figure-generation script is deposited with this paper. This paper is archived at its DOI doi:10.5281/zenodo.22645803 and its official page <https://www.kronosfusionenergy.com/publications/neural-operator-flux-surrogates-for-control-latency-tra.html>. Gyrokinetic and surrogate training run on the NVIDIA GPU HPC campaign (A100/H100/A10 class); the reduced-order layer runs on the in-house HPC node.

Data availability

This paper is archived at doi:10.5281/zenodo.22645803. The surrogate samples, the figure-generation script and the metric computations are archived with the deposit and reference the Kronos 2026 series, including the CGYRO benchmark deposit doi:10.5281/zenodo.22136279. All quantitative values trace to the register [30] by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

References

- [1] J Candy, E A Belli and R V Bravenec, A high-accuracy Eulerian gyrokinetic solver for collisional plasmas, *J. Comput. Phys.* **324** 73 (2016). doi:10.1016/j.jcp.2016.07.039
- [2] J Candy and R E Waltz, An Eulerian gyrokinetic-Maxwell solver, *J. Comput. Phys.* **186** 545 (2003). doi:10.1016/S0021-9991(03)00079-2
- [3] G M Staebler, J E Kinsey and R E Waltz, A theory-based transport model with comprehensive physics, *Phys. Plasmas* **14** 055909 (2007).
- [4] J Candy, C Holland, R E Waltz, M R Fahey and E Belli, Tokamak profile prediction using direct gyrokinetic and neutral transport simulations, *Phys. Plasmas* **16** 060704 (2009).
- [5] F Jenko, W Dorland, M Kotschenreuther and B N Rogers, Electron temperature gradient driven turbulence, *Phys. Plasmas* **7** 1904 (2000). doi:10.1063/1.874014
- [6] A J Brizard and T S Hahm, Foundations of nonlinear gyrokinetic theory, *Rev. Mod. Phys.* **79** 421 (2007).
- [7] M E Austin *et al*, Achievement of reactor-relevant performance in negative triangularity shape in the DIII-D tokamak, *Phys. Rev. Lett.* **122** 115001 (2019). doi:10.1103/PhysRevLett.122.115001
- [8] A Marinoni, O Sauter and S Coda, A brief history of negative triangularity tokamak plasmas, *Rev. Mod. Plasma Phys.* **5** 6 (2021). doi:10.1007/s41614-021-00054-0
- [9] J E Menard *et al*, Fusion nuclear science facilities and pilot plants based on the spherical tokamak, *Nucl. Fusion* **56** 106023 (2016). doi:10.1088/0029-5515/56/10/106023
- [10] B N Sorbom *et al*, ARC: a compact, high-field, fusion nuclear science facility and demonstration power plant with demountable magnets, *Fusion Eng. Des.* **100** 378 (2015). doi:10.1016/j.fusengdes.2015.07.008
- [11] A J Creely *et al*, Overview of the SPARC tokamak, *J. Plasma Phys.* **86** 865860502 (2020). doi:10.1017/S0022377820001257
- [12] L Lu, P Jin, G Pang, Z Zhang and G E Karniadakis, Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators, *Nat. Mach. Intell.* **3** 218 (2021). doi:10.1038/s42256-021-00302-5
- [13] Z Li, N Kovachki, K Azizzadenesheli, B Liu, K Bhattacharya, A Stuart and A Anandkumar, Fourier neural operator for parametric partial differential equations, *Int. Conf. on Learning Representations (ICLR)* (2021). arXiv:2010.08895
- [14] N Kovachki, Z Li, B Liu, K Azizzadenesheli, K Bhattacharya, A Stuart and A Anandkumar, Neural operator: learning maps between function spaces with applications to PDEs, *J. Mach. Learn. Res.* **24** 1 (2023).
- [15] M Raissi, P Perdikaris and G E Karniadakis, Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *J. Comput. Phys.* **378** 686 (2019). doi:10.1016/j.jcp.2018.10.045
- [16] S Wang, H Wang and P Perdikaris, Learning the solution operator of parametric partial differential equations with physics-informed DeepONets, *Sci. Adv.* **7** eabi8605 (2021).
- [17] K L van de Plassche, J Citrin, C Bourdelle *et al*, Fast modeling of turbulent transport in fusion plasmas using neural networks, *Phys. Plasmas* **27** 022310 (2020). doi:10.1063/1.5134126
- [18] J Citrin *et al*, Real-time capable first-principle based modelling of tokamak turbulent transport, *Nucl. Fusion* **55** 092001 (2015).
- [19] O Meneghini *et al*, Self-consistent core-pedestal transport simulations with neural network accelerated models, *Nucl. Fusion* **57** 086034 (2017).
- [20] O Meneghini *et al*, Integrated modeling applications for tokamak experiments with OMFIT, *Nucl. Fusion* **55** 083008 (2015). doi:10.1088/0029-5515/55/8/083008
- [21] P Rodriguez-Fernandez *et al*, Enhancing predictive capabilities in fusion burning plasmas through surrogate-based optimization of core transport solvers (PORTALS), *Nucl. Fusion* (2024).
- [22] N C Amorisco *et al*, FreeGSNKE: a Python-based dynamic free-boundary toroidal plasma equilibrium solver, *Phys. Plasmas* **31** 042517 (2024). doi:10.1063/5.0188467
- [23] L L Lao, H St John, R D Stambaugh, A G Kellman and W Pfeiffer, Reconstruction of current profile parameters and plasma shapes in tokamaks, *Nucl. Fusion* **25** 1611 (1985). doi:10.1088/0029-5515/25/11/007
- [24] F Felici *et al*, Real-time physics-model-based simulation of the current density profile in tokamak plasmas, *Nucl. Fusion* **51** 083052 (2011). doi:10.1088/0029-5515/51/8/083052
- [25] J Degraeve *et al*, Magnetic control of tokamak plasmas through deep reinforcement learning, *Nature* **602** 414 (2022). doi:10.1038/s41586-021-04301-9
- [26] M G Kapteyn, J V R Pretorius and K E Willcox, A probabilistic graphical model foundation for enabling predictive digital twins at scale, *Nat. Comput. Sci.* **1** 337 (2021). doi:10.1038/s43588-021-00069-0

- [27] A G Baydin, B A Pearlmutter, A A Radul and J M Siskind, Automatic differentiation in machine learning: a survey, *J. Mach. Learn. Res.* **18** 1 (2018).
- [28] A N Angelopoulos and S Bates, Conformal prediction: a gentle introduction, *Found. Trends Mach. Learn.* **16** 494 (2023).
- [29] K Lee, K Lee, H Lee and J Shin, A simple unified framework for detecting out-of-distribution samples and adversarial attacks, *Adv. Neural Inf. Process. Syst. (NeurIPS)* **31** (2018).
- [30] P I Ford, *Kronos Physics De-Risking Register*, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22133057
- [31] P I Ford, Hyperion as a strategic-materials platform: gain, tritium breeding, He-3 co-production, and 14 MeV neutron output from a compact spherical tokamak, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645691; official page
- [32] P I Ford, Two-code cross-validation of negative-triangularity turbulence suppression in a compact spherical-tokamak breeder: quasilinear TGLF confirms nonlinear CGYRO, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645864; official page
- [33] P I Ford, Four coupled digital twins for a compact spherical-tokamak breeder: multi-rate co-simulation, propagated uncertainty, and a real-time self-consistency monitor, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645805; official page
- [34] P I Ford, Sub-millisecond neural equilibrium reconstruction robust to sensor dropout: a virtual-sensor mesh for radiation-degraded diagnostics in a compact spherical-tokamak breeder, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645801; official page
- [35] P I Ford, Model-predictive burn control inside a certified safe envelope: a reference-governor MPC bounded by a control-barrier-function safety clamp, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645807; official page
- [36] P I Ford, Calibrated uncertainty, conformal prediction, and abstention for safety-critical fusion control: handing authority back to a deterministic floor out of distribution, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645813; official page
- [37] P I Ford, Out-of-distribution detection and epistemic gating for autonomous fusion control: a machine-readable extrapolation ledger that throttles model authority beyond the validated envelope, Kronos Fusion Energy 2026 series. doi:10.5281/zenodo.22645819; official page